

Joseph Rabara Bailey

Stanford University | josephrb@stanford.edu |

linkedin.com/in/baileyjosephr

EDUCATION

Stanford University, Stanford, CA | Expected June 2027

Coterminal B.S. in Physics and M.S. in Computer Science. Undergraduate cumulative GPA: 4.017; graduate cumulative GPA: 3.900. 159 units completed toward undergraduate degree; 27 graduate units completed.

- Selected physics and mathematics coursework: Quantum Mechanics I-II, Advanced Topics in Quantum Mechanics, Electrodynamics, Advanced Mechanics, General Relativity, Thermodynamics and Statistical Mechanics I-II, Stellar and Galactic Astrophysics, Observational Astrophysics, Probability Theory, Partial Differential Equations, Analysis, and Information Theory.
- Selected computer science coursework: Machine Learning, Natural Language Processing with Deep Learning, Machine Learning with Graphs, Deep Learning for Computer Vision, Operating Systems, Cryptography, Cryptocurrencies and Blockchain Technologies, Theory of Computation, Computer Organization and Systems, and Programming Abstractions.

Chino Hills High School, Chino Hills, CA | Class of 2023

GPA: 4.47. National Merit Finalist and ACS U.S. National Chemistry Olympiad National Finalist.

RESEARCH INTERESTS

- Scientific machine learning; AI for fundamental science; particle and neutrino physics; astronomical time-series analysis; uncertainty quantification; statistical inference; secure systems; digital assets and financial technology.

RESEARCH EXPERIENCE

Undergraduate Researcher, Stanford Physics, Prof. Benjamin Nachman | Stanford, CA | Winter 2026-Present

- Implementing and evaluating OmniFold, an unbinned machine-learning unfolding framework, on MINERvA neutrino-scattering data.
- Benchmarking machine-learning unfolding against iterative Bayesian unfolding through closure tests, generator stress tests, bootstrap resampling, covariance comparison, and bias-variance analysis.
- Developing the work toward an honors thesis chapter and manuscript outline on model dependence in neutrino cross-section extraction.

Researcher, SLAC National Accelerator Laboratory | Menlo Park, CA | June 2024-Present

- Collaborated on a four-person team proposing a particle collider based on gamma-gamma collision.
- Used Delphes, ROOT, and Unix-based high-performance computing workflows to simulate and analyze terabytes of particle-collider data.
- Applied machine-learning methods to study the effect of pileup/background particles on di-Higgs event detection.

Research Intern, Caltech LIGO Lab | Pasadena, CA | June 2023-July 2023

- Assisted with simulation, testing, and preparation of suspension elements for Laser Interferometer Gravitational-wave Observatory runs.
- Conducted experiments on laser-beam properties in an interferometer and used Python for data analysis.

RESEARCH AND TECHNICAL PROJECTS

Machine Learning-Based Unbinned Unfolding for Neutrino-Nucleus Cross Section Measurements, Stanford Physics | Stanford, CA | 2026-Present

- Testing whether high-dimensional OmniFold unfolding can reduce structural model bias relative to binned iterative Bayesian unfolding in the few-GeV neutrino regime.
- Using MINERvA open data and simulated pseudo-data to compare model-induced bias, correlation preservation, statistical coverage, and uncertainty stability.

Physically Motivated Time-Series Photometry of AM CVn, PHYSICS 100: Introduction to Observational Astrophysics | Stanford, CA | Spring 2026

- Reduced 140 V-band CCD images with bias, dark, and star-masked flat-field calibration; performed AAVSO-sequence differential photometry on 138 retained exposures.
- Recovered AM CVn variability consistent with the known 1051 s positive-superhump/harmonic family and evaluated whether a 111 minute baseline could refine published period constraints.

Video-Based Person Re-Identification with LoRA-Adapted Vision Foundation Models, CS 231N: Deep Learning for Computer Vision | Stanford, CA | Spring 2026

- Built a video person re-identification ablation framework using DINOv2/DINOv3 vision backbones, LoRA adaptation, BNNeck training, triplet loss, and cosine retrieval.
- Compared frozen encoders, LoRA-adapted frame encoders, mean pooling, and lightweight temporal attention on the MARS benchmark.

SPLoRA: Parameter-Efficient Fine-Tuning of GPT-2 with LoRA, CS 224N: Natural Language Processing with Deep Learning | Stanford, CA | Winter 2026

- Implemented LoRA for GPT-2 for Quora paraphrase detection and Shakespearean sonnet generation.
- Ran ablations over adapter placement and hyperparameters; achieved 0.888 development accuracy while training roughly 1% of model parameters and 41.94 chrF on sonnet generation.

Soccer Action Prediction with Graph Neural Networks, CS 224W: Machine Learning with Graphs | Stanford, CA | Autumn 2025

- Modeled soccer possessions as heterogeneous graphs using Wyscout data and trained GraphSAGE and Relational Graph Transformer models with RelBench and PyG to predict best next action.

Gaussian Process Bump Hunting, Jefferson Lab codebase | 2025

- Developed Gaussian-process regression code to search for bumps/excesses associated with the hypothetical X17 particle in particle-collider data.
- Presented results at the APS Far West Section Annual Meeting at UC Santa Cruz.

On-Chain Ethereum Wallet, 2025

- Built a programmable Solidity and Foundry wallet supporting ETH and ERC-20 transfers, approvals, deposits, withdrawals, and a 2-of-3 multisig authorization scheme.

Pintos OS Enhancements, 2025

- Implemented thread scheduling, system calls, user programs, virtual memory, and file-system support for a primitive x86 operating system using C and kernel programming.

End-to-End Encrypted Chat Client, 2025

- Implemented the Double Ratchet Algorithm, based on protocols used in Signal and WhatsApp, for secure messaging.

PRESENTATIONS AND MANUSCRIPTS IN PREPARATION

- Presenter, Gaussian Process Bump Hunting, APS Far West Section Annual Meeting, University of California, Santa Cruz, 2025.
- Honors thesis and manuscript in preparation: systematic evaluation of OmniFold in the few-GeV neutrino regime using MINERvA open data.

PROFESSIONAL AND ENGINEERING EXPERIENCE

Incoming Product Management Intern, Digital Assets Team, WisdomTree Digital Movement, Inc. | Austin, TX | Summer 2026

- Accepted offer to support product management, market research, internal documentation, and product-development discussions for digital-asset financial products.

Engineering Intern, Polytec | Pasadena, CA | June 2023-Sept. 2023

- Designed statistical tools for customers analyzing vibration data with Python, MATLAB, and Octave.
- Modeled and 3D-printed laboratory tools to streamline operations.

HONORS AND AWARDS

- Stanford Track and Field Heart and Hustle Award, 2025.
- National Merit Finalist.
- ACS U.S. National Chemistry Olympiad National Finalist.
- Music Teachers' Association of California Certificate of Merit, Piano Level 10; State Honors and Branch Honors.
- Chino Hills High School: Best All-Around Male Student and Homecoming King.
- Black Belt in Karate.

LEADERSHIP, ATHLETICS, SERVICE, AND MUSIC

Student-Athlete, Stanford Track and Field, NCAA Division I | Stanford, CA | Jan. 2024-Present

- Hurdler and sprinter, including the 400-meter hurdles, training more than 20 hours per week while carrying a demanding Physics and Computer Science course load.

Dorm Co-President, Stanford freshman dorm | Stanford, CA | Sept. 2023-June 2024

- Helped lead a 90-person dorm community through weekly meetings, event planning, peer engagement, and community standards.

Musician, Stanford Jazz Orchestra and Stanford Jazz Combos | Stanford, CA | Sept. 2023-Jan. 2024

- Performed alto saxophone, flute, and clarinet in Stanford jazz ensembles.

Section Leader, Jazz Captain, and Saxophone Soloist, Chino Hills High School Marching Band, Drumline, and Jazz Band | Chino Hills, CA | Aug. 2019-Apr. 2023

- Led alto saxophone sectionals, served as jazz captain, and performed as marching-band saxophone soloist for three championship seasons.

TECHNICAL SKILLS

Programming: Python, C/C++, Java, JavaScript, Solidity, MATLAB, Octave

Machine Learning: PyTorch, LoRA, graph neural networks, GraphSAGE, Relational Graph Transformers, OmniFold, Gaussian processes, boosted decision trees, neural networks

Physics and Data Analysis: ROOT, Delphes, MINERvA open data, detector simulation, CCD photometry, periodograms, differential photometry, covariance analysis, high-performance Unix computing

Systems and Security: Operating systems, kernel programming, cryptography, AES-GCM, SHA-256, Double Ratchet, Ethereum smart contracts, Foundry

Languages: First-year Modern Chinese coursework